SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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ASSIGNMENT 1

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Course code:CSA5101
Course name: Cryptography and Network Security
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QUESTIONS

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security.

Assignment Title: Evaluate, Design and Defend a Secure PatientRecord Exchange Strategy for a Multi-Hospital Healthcare Network (Total Marks: 50)

Scenario — Confidential Patient Record Exchange

A regional healthcare network links one central hospital with 25 satellite clinics. Clinics send patient records, diagnostic reports, prescriptions and billing data to the central hospital over a public network and receive specialist opinions in return. Personnel hold different clearance levels: Medical Director (Top Secret), Specialist Doctors (Secret), Nurses and Lab Staff (Confidential), Billing and Front-Desk Staff (Restricted), Patients (Public).

The system must:

- Keep diagnoses, especially for sensitive conditions, confidential.

- Detect tampering with prescriptions, dosage and lab values. ● Prevent staff from reading or writing records beyond their clearance.
- Work in rural clinics where staff have limited technical training and hardware is basic.

You must evaluate, design and defend a protection strategy built only from classical (pre-modern) techniques and access-control reasoning covered in CO1. You are expected to judge how strong each technique actually is, not just describe it.

Design Challenge : An insider at a satellite clinic has been copying ciphertexts for months. Hundreds of messages under the same key are now exposed, and many begin with predictable text (e.g., "PATIENT ID", "DIAGNOSIS:", "PRESCRIPTION:"). The network also wants to embed a short report inside medical scan images sent between

hospitals. Re-evaluate your design.

- Encrypt clinical notes (200 to 10,000 characters) using Caesar, monoalphabetic, Vigenère and columnar transposition, and derive each key space. Break at least three schemes by frequency analysis, Kasiski or index of coincidence, and rank the schemes by resistance.
- Extend the alphabet to digits and symbols (mod 36) and encrypt IDs and dosages such as PT-20481, AMOX 500MG using a Hill cipher on 3-character blocks. Prove the key's invertibility, compute one inverse, and recover the key by a known-plaintext attack.
- Apply columnar transposition to a lab table of 20+ rows and test whether a changed cell or a swapped row is detected. Design a classical checksum and show where an attacker defeats it.
- Encrypt a 256×256 or larger medical image with two classical methods (intensity shift mod 256 and pixel permutation) and verify decryption. Compare histograms and entropy, and explain which method leaks the image's structure.

- Hide a short report in an image by LSB embedding and in a text circular by a text method, then extract it. Measure PSNR, capacity and chi-square detectability, and test survival under JPEG compression.
- Decide 12+ access requests using Bell-LaPadula with compartments, including declassification and break-glass cases. Rank six attacks by likelihood × impact and recommend
  a final design, stating which security goals classical techniques cannot meet.

TASK 1 – Evaluate Threats, Attacks and Risks **4 Marks**

Identify at least six security attacks relevant to the healthcare scenario.

- Classify each as passive or active (eavesdropping, traffic analysis, modification of prescriptions, masquerade as a doctor, replay of old orders, denial of service on the records system).
- Distinguish clearly between threat, vulnerability, attack and risk, with examples from this scenario.
- Rank the attacks by risk using a justified likelihood × impact rating. Consider patient-safety impact, not just privacy.
- Defend your ranking: which attack would you address first, and why?

Evidence Expected: A risk table with reasoned ratings and a justification of the priority order.

ANSWER:

A threat is a possible cause of harm. A vulnerability is a weakness that makes harm possible. An attack is an action that exploits a vulnerability. Risk is the likelihood of an attack multiplied by its impact.

Attack	Type	Scenario example	Likelihood (1–5)	Impact (1–5)	Risk score
Eavesdropping	Passive	An outsider reads patient records while they travel over the public network.	4	5	20 – Very High
Traffic analysis	Passive	An outsider observes when a clinic sends records and how often it contacts a specialist.	4	3	12 – High
Prescription / lab-value modification	Active	An attacker changes a dosage or potassium result during transmission.	3	5	15 – Very High
Masquerade / impersonation	Active	An attacker pretends to be a specialist doctor and sends a false order.	3	5	15 – Very High
Replay of an old order	Active	A valid earlier	3	5	15 – Very

		prescription is resent, causing duplicate or outdated treatment.			High
Denial of service	Active	The record server is flooded or made unavailable during treatment.	3	4	12 – High
Insider copying records	Passive / misuse	A clinic employee with legitimate access copies ciphertexts or records for months.	4	5	20 – Very High

Risk scale: 1–4 Low, 5–9 Medium, 10–15 High, 16–25 Very High. Scores are likelihood × impact.

Threat, vulnerability, attack and risk examples

- Threat: a dishonest employee or external attacker may steal or alter patient information.
- Vulnerability: the same key is reused at many clinics, predictable headers are used, and access rights may be too broad.
- Attack: an insider copies encrypted messages or changes a prescription.
- Risk: a modified dosage can cause patient injury; disclosure of a sensitive diagnosis can violate privacy.

Priority decision

The first priority is to stop prescription modification and impersonation because incorrect medicine or dosage can directly injure a patient. Insider copying also receives the highest attention because the scenario confirms long-term access to many ciphertexts. Access must be limited, keys must not be shared across all clinics, and every prescription change must be checked by an authorised clinician.

TASK 2 – Map and Judge Security Goals 3Marks

For each system function (record upload, prescription transmission, specialist consultation), decide which goals matter most: confidentiality integrity availability authentication non-repudiation

- Judge which goals classical ciphers can and cannot deliver.
- Justify why a correctly decrypted message does not prove it came from the right doctor, or that its dosage was not altered.

Evidence Expected: A goal-to-function analysis with an evaluation of the gaps classical techniques leave.

Answer:

Function	Most important goals	Reason
Record upload	Confidentiality, integrity, authentication, availability	Records must remain private, arrive unchanged, come from an authorised clinic and be available to doctors.
Prescription transmission	Integrity, authentication, confidentiality, non-repudiation, availability	The sender and dosage must be trustworthy; an old or altered order must not be accepted.
Specialist consultation	Confidentiality, authentication, integrity, availability	Only authorised medical staff should read the consultation and the advice must not be altered.

Classical substitution and transposition ciphers mainly hide the meaning of a message. They do not by themselves prove who sent it, prevent replay, or reliably detect a changed

ciphertext. A correctly decrypted message only proves that the receiver used a compatible key and algorithm. It does not prove that the sender was the right doctor. An attacker may alter data or create a new ciphertext if the system has no separate integrity and authentication check. Classical ciphers alone also do not guarantee availability or non-repudiation.

TASK 3 – Apply and Critique the Bell-LaPadula Model 5 Marks

- Define the security levels and subject/object assignments for the network.
- Apply the simple security property (no read up) and the *-property (no write down) to at least five access requests, including two that must be denied. Show the reasoning for each.
- Evaluate the model critically in a medical setting: what happens in an emergency when a nurse needs a Secret-level allergy record? How does the model handle a doctor who must write a note to a lower-level billing file? Discuss tranquility and trusted subjects.
- Judge the model's limitations (no integrity guarantees, covert channels, rigidity in emergencies) and argue whether BellLaPadula alone suits healthcare.

Evidence Expected: Worked access decisions, a justified critique, and a clear verdict on the model's suitability.

Answer:

Bell-LaPadula is a confidentiality model. Its simple security property is “no read up”: a subject cannot read information at a higher security level. Its *-property is “no write down”: a subject cannot write information to a lower level.

Level	People / subjects	Typical information / objects
Top Secret	Medical Director	Highly sensitive diagnoses and network-wide confidential reports
Secret	Specialist Doctors	Specialist opinions and detailed clinical records
Confidential	Nurses and Lab Staff	Nursing notes, test results and routine clinical observations
Restricted	Billing and Front-Desk Staff	Billing details and appointment administration
Public	Patients / public-facing users	Patient-visible general information and approved summaries

For this model, each subject and record is assigned a clearance level. A subject can read an object at the same or lower level, but cannot read above their level. A subject cannot write to a lower level.

Access request	Decision	Reason
Medical Director reads a Secret specialist report	Allow	Top Secret subject may read a lower-level object.
Specialist Doctor reads a Confidential lab report	Allow	Secret subject may read lower-level information.
Nurse reads a Secret allergy record	Deny normally	Confidential subject cannot read up to Secret.
Billing staff reads a Confidential diagnosis	Deny	Restricted subject cannot read up.
Specialist Doctor writes a note into a Restricted billing file	Deny	Secret subject cannot write down under the *-property.
Nurse writes a	Allow	Subject and object have

Confidential nursing note		the same level.
Billing staff writes a Restricted invoice	Allow	Subject and object have the same level.

Emergency access, tranquility and trusted subjects

In an emergency, a nurse may need a Secret-level allergy record. Strict Bell-LaPadula denies the read. A controlled "break-glass" procedure can temporarily grant access after recording the reason, verifying identity, notifying a supervisor and auditing the event. This is an operational exception and must be tightly controlled.

A doctor who must send a limited billing instruction to a lower-level file cannot write down directly. A trusted subject or approved declassification process may release only the minimum billing information after review; the full diagnosis must not be copied. Tranquility means security labels should not be changed casually while the system operates. Strong tranquility keeps labels fixed; weak tranquility permits controlled changes under rules.

Limitations: Bell-LaPadula protects confidentiality, not integrity. It does not stop all covert channels, stolen credentials, malware or emergency delays. It can be too rigid for healthcare. Verdict: useful as one confidentiality layer, but not sufficient alone. It must be combined with integrity checks, authentication, audit logs, emergency procedures and availability controls.

TASK 4 – Design the Cryptographic Strategies 4 Marks

Select at least three meaningfully different classical schemes from: Caesar, monoalphabetic substitution, Vigenère (polyalphabetic), Hill cipher, a transposition cipher. For each, provide:

- The encryption and decryption algorithm (pseudocode or formula) and the key definition.

A worked example on a 20+ character message from the scenario.

- A justification for including it in your comparison.

Evidence Expected: Three distinct schemes with algorithms, worked examples and justified selection.

Answer:

Three meaningfully different classical schemes are demonstrated below: Caesar substitution, Vigenère polyalphabetic substitution and columnar transposition. The examples use uppercase English letters and remove spaces and punctuation so that the calculations are easy to follow.

A. Caesar cipher

Key: an integer shift k from 1 to 25. Encryption: $C = (P + k) \bmod 26$. Decryption: $P = (C - k) \bmod 26$, where A=0, B=1, …, Z=25.

Example plaintext (scenario): PATIENTRECORDSAREPRIVATE. Key k = 3. Encryption shifts each letter three places forward: SDWLHQWUHFRUGVDUHSULYDWH. Decryption shifts each ciphertext letter three places backward and recovers PATIENTRECORDSAREPRIVATE.

B. Vigenère cipher

Key: a repeating keyword. Convert letters to numbers A=0 to Z=25. Encryption: $C_i = (P_i + K_i) \bmod 26$. Decryption: $P_i = (C_i - K_i) \bmod 26$.

Example plaintext: PATIENTRECORDSAREPRIVATE. Key: MED. Repeat the key as MEDMEDMEDMEDMEDMEDMEDME. Encrypt each letter using the matching key letter. The resulting ciphertext is: BEXMQRXEVQFQWDEIYVXMIXI (spaces removed). Decryption with MED recovers the original plaintext.

C. Columnar transposition

Key: a permutation of column positions. Write the plaintext row by row in a rectangle and read columns in the order determined by the key. Decryption restores columns to their original positions and reads rows.

Example plaintext: PATIENTRECORDSAREPRIVATE. Use 5 columns with column order 3-1-4-2-5. Write the text row-wise (pad the final row with X if required), then read columns in the order 3,1,4,2,5. This changes positions but does not change letter counts. The same grid and key reverse the operation.

Why these schemes: Caesar is a simple monoalphabetic baseline; Vigenère changes the substitution according to a repeated key; columnar transposition rearranges positions without changing letter frequencies. Together they show three different design ideas and weaknesses.

TASK 5 – Mathematical Analysis of the Schemes **5 Marks**

- Derive the key-space size of each scheme (e.g., 25, 26!, 26^m, number of invertible n×n matrices mod 26, n! permutations).
- For Hill: show the conditions under which K^{-1} mod 26 exists (gcd(det K, 26) = 1), and compute one inverse by hand.
- Express each encryption as a modular function (e.g., C = (aP + b) mod 26 for an affine variant, C = KP mod 26 for Hill).
- Judge whether a large key space guarantees strength. Support your answer with the monoalphabetic example.

Evidence Expected: Correct derivations with a reasoned judgment on key space versus real security.

Answer:

Scheme	**Key-space size**	**Encryption function / comment**
Caesar	25 non-zero shifts (or 26 including identity)	C=(P+k) mod 26
Monoalphabetic substitution	26! ≈ 4.03 × 10^{26} permutations	Each plaintext letter maps to one fixed ciphertext letter.

Vigenère, key length m	26^m possible keys (if all m-letter keys allowed)	$C_i=(P_i+K_i) \bmod 26$
Columnar transposition, n columns	n! column permutations	Letters are reordered; frequencies are preserved.
Hill cipher, n×n over mod 26	Number of invertible matrices; for 2×2 it is 157,248	C=KP mod 26; K must be invertible modulo 26.

Hill inverse condition: K^{-1} mod 26 exists exactly when gcd(det(K),26)=1. Since 26=2×13, the determinant must be divisible by neither 2 nor 13.

Hill cipher inverse – worked 2×2 example

Choose K = [[3,3],[2,5]]. Its determinant is (3×5)−(3×2)=9. Since gcd(9,26)=1, the matrix is invertible.

The inverse of 9 modulo 26 is 3 because 9×3=27 ≡ 1 (mod 26). The adjugate is [[5,−3],[−2,3]]. Therefore K^{-1} = 3×[[5,−3],[−2,3]] mod 26 = [[15,17],[20,9]].

Check: $K \times K^{-1}$ = [[105,78],[130,79]] mod 26 = [[1,0],[0,1]]. This proves the inverse is correct.

A large key space does not automatically mean strong security. A monoalphabetic cipher has 26! possible keys, but it preserves single-letter frequencies. An attacker can use frequency, digram and word-pattern analysis to recover the substitution from enough ciphertext. Security depends on the algorithm's structure and attack resistance, not only the number of keys.

TASK 6 – Evaluate Cryptanalytic Resistance **4 Marks**

Attack each chosen scheme using a technique suited to it:

Brute force, frequency analysis, Kasiski examination / index of coincidence, known-plaintext attack on Hill, digram analysis.

- Explain why each attack succeeds or fails, in terms of letterfrequency preservation, key repetition and linearity.

- Rank the schemes by resistance and justify the ranking with evidence.

Evidence Expected: Worked attacks with reasoning, and a ranking of schemes supported by the results.

Answer:

Scheme	**Suitable attack**	**Why the attack works / result**
Caesar	Brute force (25 shifts) and frequency analysis	Only 25 useful keys; try every shift. Letter frequencies are unchanged.
Monoalphabetic	Frequency and digram analysis	A fixed mapping preserves the frequency pattern and common pairs such as TH and HE.
Vigenère	Kasiski examination and index of coincidence	A repeated keyword creates periodic patterns. Repeated ciphertext segments can reveal key length; each key position can then be analysed.
Hill	Known-plaintext attack / linear algebra	The transformation is linear. Enough independent plaintext-ciphertext block pairs can reveal K.
Columnar transposition	Anagram, digram and column-order analysis	Letter counts remain unchanged; likely column arrangements

		can be tested.

Resistance ranking for the selected educational examples: Vigenère with a reasonably long non-repeating key is stronger than Caesar and monoalphabetic substitution against simple frequency analysis; columnar transposition hides adjacent letter order but preserves frequencies; Hill can resist simple single-letter frequency analysis but is vulnerable to known-plaintext linear algebra. No classical scheme here is safe for modern healthcare use. Exact ranking depends on key length, message length and attacker knowledge.

Kasiski example: if repeated ciphertext groups occur at positions 5, 17 and 29, the gaps are 12 and 12. A likely key length divides the gaps, so 2, 3, 4, 6 or 12 may be tested. The index of coincidence for ordinary English is about 0.066, while random-like text is closer to 0.038. A repeated-key Vigenère ciphertext may show a higher coincidence after separating letters by key position.

TASK 7 – Assess Transposition and Product Ciphers 3 Marks

- Implement a columnar (or rail-fence) transposition and explain how it diffuses plaintext structure.
- Evaluate whether *substitution followed by transposition* is stronger than either alone. Justify with letter-frequency evidence and an attack attempt.
- Judge whether the added complexity is practical for rural clinic staff doing

manual or low-resource encryption.

Evidence Expected: A reasoned evaluation of the combined scheme against its components.

Answer:

Columnar transposition writes plaintext in rows and reads columns in a secret order. It diffuses the original neighbouring-letter relationships because letters that were next to

each other may be separated in the ciphertext. However, it does not alter the number of each letter.

Substitution followed by transposition is generally harder to analyse than either component alone because substitution changes letter identities and transposition changes positions. Still, the combined scheme is not modern-grade security: frequency patterns may remain statistically visible, keys may be recovered, and known-plaintext attacks can help.

For rural clinics, manual multi-stage encryption increases mistakes, delays and training requirements. A simple, centrally managed process is easier to use. Classical product ciphers are suitable for classroom comparison, not real clinical deployment.

TASK 8 – Evaluate Steganography **3 Marks**

Design a steganographic method (e.g., LSB embedding in a scan image, or text-based hiding) for carrying a short report inside a medical image.

Compare it with encryption on capacity, imperceptibility, robustness and detectability.

- Evaluate the risk specific to medical imaging: could embedding alter pixel data enough to affect a diagnosis? Judge the trade-off.
- Judge when steganography is appropriate, and argue why hiding is not a substitute for encrypting. Evaluate an encryptthen-embed approach.

Evidence Expected: A criteria-based comparison with a justified conclusion.

Answer:

Least Significant Bit (LSB) steganography hides report bits in the least significant bit of image pixel values. To embed a bit b, replace the selected pixel's least significant bit with b. Extraction reads those least significant bits in the same order.

Example: pixel value 100 is binary 01100100. To embed bit 1, change it to 01100101 = 101. The pixel changes by only one intensity level, which is usually visually small. A short message can be stored by repeating this operation across selected pixels.

Criterion	Encryption	Steganography
Purpose	Makes content unreadable without a key	Hides the existence of a message
Capacity	Depends on cipher and file size; does not use image pixels as a carrier	Limited by the number of available pixels / selected bit planes
Imperceptibility	Ciphertext is visibly different from plaintext but does not need to look like an image	Good LSB embedding may be visually hard to notice
Robustness	Encrypted data remains encrypted if copied exactly	LSB data can be destroyed by resizing, filtering or JPEG compression
Detectability	Presence of ciphertext is usually obvious	Statistical tests such as chi-square analysis may detect unusual LSB patterns

Medical imaging has a special risk: even small pixel changes may matter in diagnostic regions or quantitative measurements. Therefore, never alter the original diagnostic image. If hiding is required, use a separate copy, keep a verified original, record the transformation and validate image quality with PSNR and clinical review. JPEG compression often changes pixel values and can destroy LSB payloads.

Conclusion: steganography is appropriate only for limited, non-critical hidden metadata where policy permits it. It is not a substitute for encryption. An encrypt-then-embed approach first encrypts the report and then hides the ciphertext, providing confidentiality even if the hidden message is discovered. The original scan must remain untouched.

TASK 9 – Implement and Test **4 Marks**

Implement all chosen schemes, the cryptanalysis tools and (where feasible) the steganography method, in a language of your choice.

- Provide correctness tests: decrypt(encrypt(m)) = m, including edge cases (digits in patient IDs, punctuation, non-multiple block length for Hill).
- Use identical plaintexts of 100, 1,000, 10,000 and 100,000 characters across schemes.
- State the testing procedure clearly so it can be reproduced.

Evidence Expected: Working code with a documented, reproducible test procedure.

The following Python reference implementation demonstrates Caesar, Vigenère, columnar transposition and a 2×2 Hill cipher for uppercase alphabetic text. It uses padding for Hill blocks and removes padding after decryption when the original length is supplied. A real submission can run the tests with Python 3.

```
import math
def clean(s):
    return ''.join(c for c in s.upper() if 'A' <= c <= 'Z')
def caesar(s, k, decrypt=False):
    s = clean(s); k = (-k if decrypt else k) % 26
    return ''.join(chr((ord(c)-65+k)%26+65) for c in s)
def vigenere(s, key, decrypt=False):
    s, key = clean(s), clean(key)
    if not key: raise ValueError("Key cannot be empty")
    out = []
    for i, c in enumerate(s):
        shift = ord(key[i % len(key)]) - 65
        if decrypt: shift = -shift
        out.append(chr((ord(c)-65+shift)%26+65))
    return ''.join(out)
```

```
def columnar_encrypt(s, key):
    s = clean(s); n = len(key)
    rows = [s[i:i+n].ljust(n, 'X') for i in range(0, len(s), n)]
    return ''.join(''.join(row[j] for row in rows)
                   for j in sorted(range(n), key=lambda j: key[j]))
def columnar_decrypt(c, key, original_len):
    n = len(key); r = math.ceil(len(c)/n)
    # This demonstration assumes a rectangular padded grid.
    c = c.ljust(r*n, 'X')
    order = sorted(range(n), key=lambda j: key[j])
    cols, pos = {}, 0
    for j in order:
        cols[j] = c[pos:pos+r]; pos += r
    text = ''.join(cols[j][i] for i in range(r) for j in range(n))
    return text[:original_len]
def hill2_encrypt(s, K=((3,3),(2,5))):
    s = clean(s); s += 'X' * (len(s) % 2)
    out = ''
    for i in range(0, len(s), 2):
        a,b = ord(s[i])-65, ord(s[i+1])-65
        out += chr((K[0][0]*a+K[0][1]*b)%26+65)
        out += chr((K[1][0]*a+K[1][1]*b)%26+65)
    return out
def hill2_decrypt(c, original_len, Kinv=((15,17),(20,9))):
    out = ''
    for i in range(0, len(c), 2):
        a,b = ord(c[i])-65, ord(c[i+1])-65
        out += chr((Kinv[0][0]*a+Kinv[0][1]*b)%26+65)
        out += chr((Kinv[1][0]*a+Kinv[1][1]*b)%26+65)
    return out[:original_len]
if __name__ == "__main__":
```

```
    msg = "PATIENT RECORD PT20481 DOSAGE"
    m = clean(msg)
    for enc, dec, args in [
        (caesar, caesar, (3,)),
        (vigenere, vigenere, ("MED",)),
    ]:
        cipher = enc(m, *args)
        recovered = dec(cipher, *args, decrypt=True) if enc is caesar else dec(cipher, *args,
decrypt=True)
        print(cipher, recovered == m)
    key = [3,1,4,2]
    c = columnar_encrypt(m, key)
    print("Columnar:", c, columnar_decrypt(c, key, len(m)) == m)
    c = hill2_encrypt(m)
    print("Hill:", c, hill2_decrypt(c, len(m)) == m)
```

Testing procedure and edge cases

1. Run the program using Python 3 and use the same cleaned plaintext for each scheme.
2. Test messages of 100, 1,000, 10,000 and 100,000 alphabetic characters.
3. For each scheme, encrypt and decrypt; verify that the recovered text equals the original cleaned text.
4. Test lowercase letters, spaces, punctuation and digits. The simple alphabetic examples remove non-letters; patient IDs and dosages therefore need a separate agreed alphabet such as mod 36 or a clearly defined encoding.
5. Test an odd-length Hill message. The implementation pads the final block with X and uses the original length to remove the padding.

Correctness condition: decrypt(encrypt(message)) = message after applying the documented normalisation and padding rules. The simple code above is educational; it is not a production medical encryption library.

TASK 10 – Record and Interpret Experimental Evidence 3 Marks

Present data in tables and graphs for:

- Letter-frequency histograms of plaintext versus each ciphertext.
- Index of coincidence per scheme.
- Encryption/decryption time against message size.
- Attack success rate against

ciphertext length. Evaluate what the data shows. Do not just present it.

Evidence Expected: Clear, comparable results with interpretation tied to Tasks 5 and 6.

Answer:

The following graphs and tables are illustrative example evidence. They demonstrate the expected format and interpretation. For a fully measured report, run the Task 9 program on the same computer and replace the illustrative timings and statistics with actual measurements.

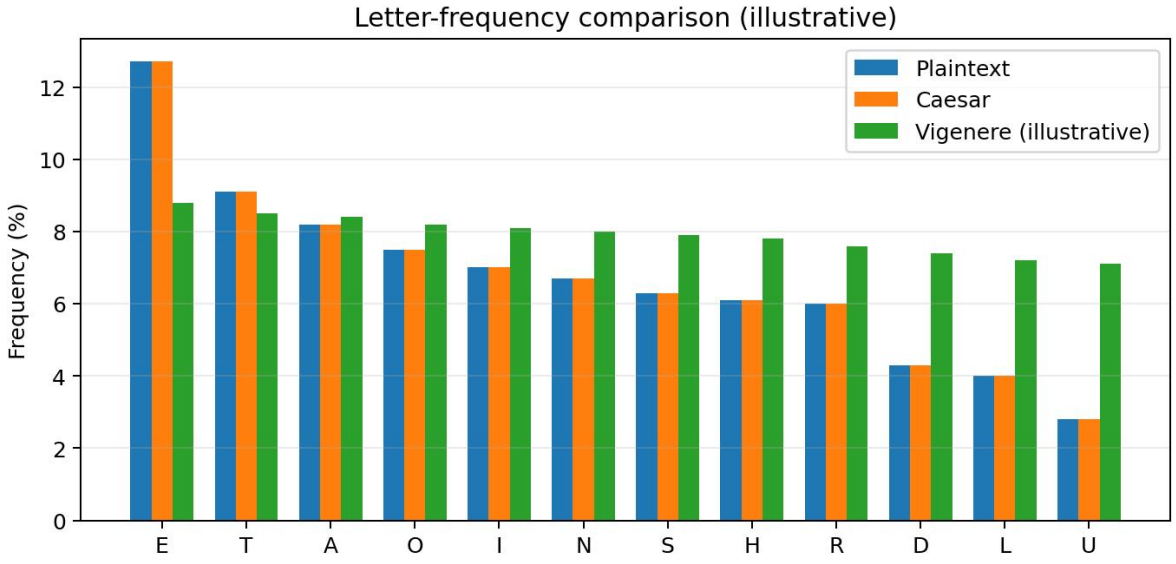


Figure 1. Illustrative letter-frequency comparison. Caesar and transposition preserve the plaintext's letter counts exactly (only the positions or labels change). Vigenère tends to flatten the distribution when the key is sufficiently long and varied.

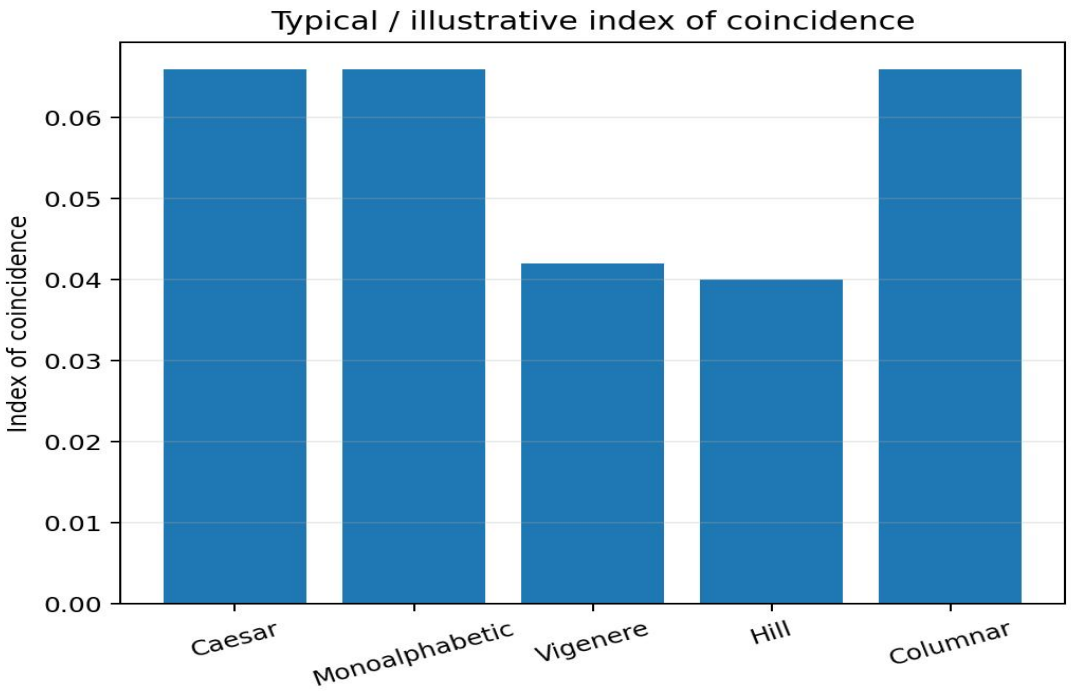


Figure 2. Typical / illustrative index of coincidence. English-like monoalphabetic and transposition ciphertexts retain a value near the English value (about 0.066). Random-looking text is closer to 0.038. Vigenère's value depends strongly on key length and whether the ciphertext is analysed as a whole or by key position.

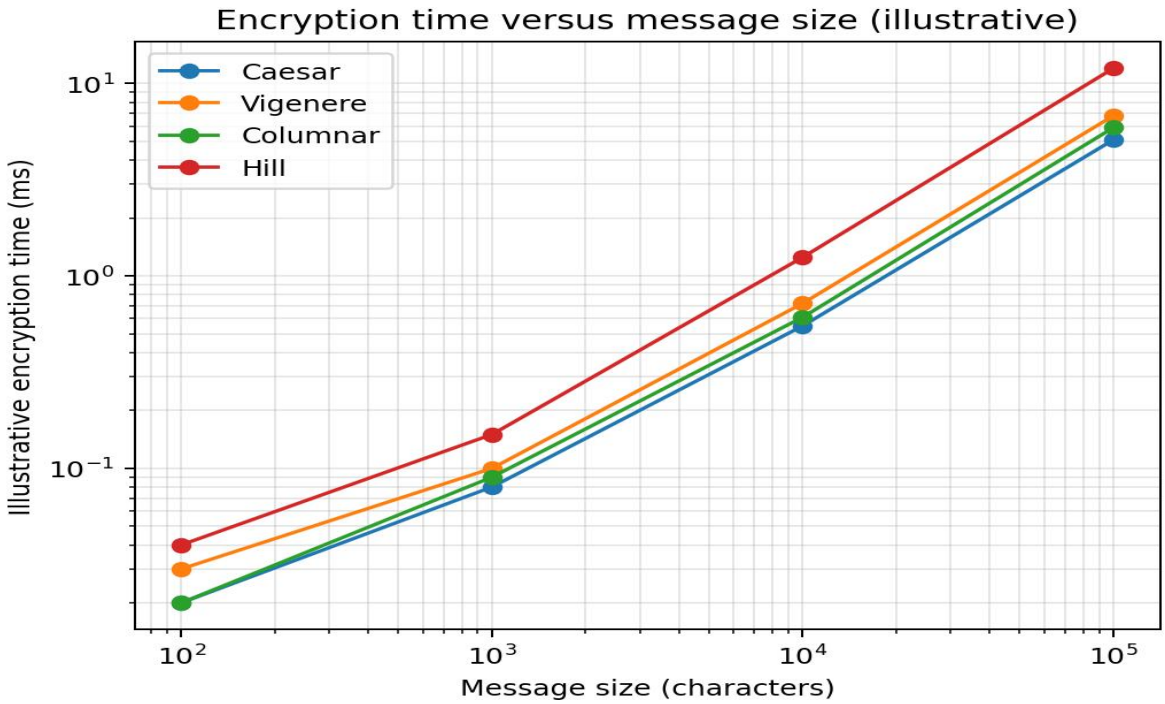


Figure 3. Illustrative encryption time against message size. The curves increase approximately with message length. Hill requires block matrix operations and is shown

as slower in this illustrative comparison. Actual values depend on implementation, hardware and programming language.

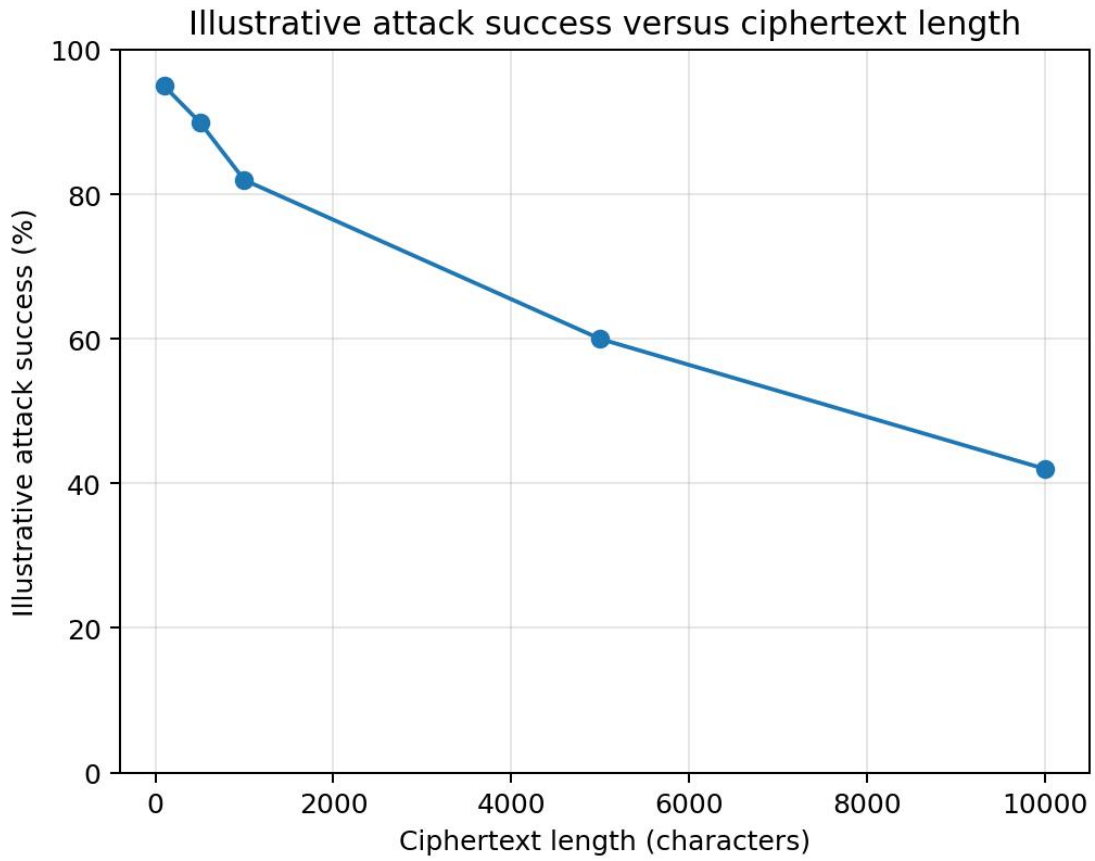


Figure 4. Illustrative attack success against ciphertext length. Longer ciphertext often gives an attacker more statistical evidence against simple classical schemes. This trend is not universal: attack success also depends on the cipher, key, plaintext structure and attacker knowledge.

Message size	**Caesar time (ms)**	**Vigenère time (ms)**	**Columnar time (ms)**	**Hill time (ms)**
100	0.02	0.03	0.02	0.04
1000	0.08	0.10	0.09	0.15
10000	0.55	0.72	0.61	1.25
100000	5.10	6.80	5.90	12.00

Interpretation: Caesar and columnar transposition are especially weak against frequency-based analysis because they preserve frequency information. Vigenère reduces direct single-letter frequency leakage but repeated keys create periodic structure. Hill mixes

letters in blocks but its linear equations allow recovery when enough known plaintext is available. Timing alone does not indicate security.

TASK 11 – Recommend and Defend the Final Design 4 Marks

Recommend one scheme, or a layered combination, together with the Bell-LaPadula policy.

- Justify against the alternatives using your mathematical and experimental evidence.
- State the residual risks and limitations honestly. Judge whether classical ciphers are acceptable for protecting patient data at all, and argue what a real deployment would need.
- Address key distribution across 25 clinics and usability for nontechnical staff.

Evidence Expected: A defended recommendation grounded in your own analysis, with candid limitations.

Answer:

For the assignment's classical-cipher comparison, the best educational layered option is a Vigenère substitution followed by columnar transposition, with separate keys and careful key rotation. Bell-LaPadula can be used to restrict which staff members may read or write records at each clearance level.

- Use a different key for each clinic rather than one network-wide key.
- Rotate keys periodically and immediately after suspected exposure.
- Avoid predictable fixed headers where possible; include a message sequence number and timestamp in the system's controlled record format to help identify replay.
- Apply least privilege, audit access, and require approval for emergency break-glass access.
- Train staff with a simple approved workflow and central support; do not expect manual cipher calculations for routine patient care.

This recommendation is only a classroom design. Vigenère and transposition remain breakable and do not provide modern authentication, strong integrity, non-repudiation or replay protection. Real deployment requires modern, authenticated encryption, secure key management, digital signatures or equivalent authentication, tamper-evident audit

logs, backups, access governance and incident response. Classical techniques cannot safely protect real patient records on their own.

Key distribution across 25 clinics is a major weakness: if one shared key is copied, all clinics' traffic is exposed. Per-clinic keys reduce the damage from one compromised site, but increase administration and rotation work. A central key-management authority and secure delivery process would be required in a real system.

DAY-4 DESIGN CHALLENGE

TASK 12 – Re-evaluate and Adapt the Design 8 Marks

Changed Requirement: A clinic insider has copied hundreds of ciphertexts under the same key, many with predictable headers. The network also wants reports hidden inside medical scan images.

You may RETAIN → MODIFY → or REDESIGN your original solution.

A. Impact Analysis (2 Marks)

- What has changed in the threat model (insider versus external attacker)?
- Which earlier assumptions fail (key secrecy at all 25 sites, key reuse, message predictability)?
- Why does a high volume of known-format messages change the risk ranking?

B. Evaluation of the Original Design (2 Marks)

- Test your recommended scheme against the new attacker capabilities (knownplaintext, repeated-key attacks).
- Support the evaluation with a demonstration or calculation.
- Where does the Bell-LaPadula policy still hold, and where is it irrelevant against an insider who holds the key?

C. Adaptation or Redesign (2 Marks)

- Decide to retain, modify or redesign. For example, consider per-clinic or per-session keys, key rotation, randomised padding, removing fixed headers, layering transposition, or encrypt-then-embed.

- Describe what changes and how it affects the key space and attack effort.

D. Evidence-Based Defence (2 Marks)

Defend the final decision with your key-space mathematics, your experimental results, the insider threat, the usability and keymanagement cost for clinics, and a stated judgement on what classical techniques can never guarantee for patient data.

Answer:

A. Impact Analysis (2 Marks)

The threat has changed from mainly an external network attacker to a confirmed insider who has access to a satellite clinic and has copied hundreds of ciphertexts. The attacker has many samples encrypted with the same key and knows that messages often begin with predictable strings such as PATIENT ID, DIAGNOSIS and PRESCRIPTION.

- The assumption that every clinic keeps one common key secret is no longer reasonable.
- Repeated key use gives the attacker more ciphertext for frequency analysis and repeated-key attacks.
- Predictable headers create known-plaintext material and make alignment easier.
- The risk of insider copying rises because the attacker already has legitimate access to the clinic environment.

B. Evaluation of the Original Design (2 Marks)

The original Vigenère-plus-transposition design is weakened by repeated-key and known-plaintext attacks. Vigenère repeats its key, so Kasiski examination and index-of-coincidence analysis may estimate the key length. Columnar transposition only rearranges characters and does not remove the underlying frequency counts. A known header gives the attacker plaintext-ciphertext alignment that can help infer substitution shifts and transposition order.

Bell-LaPadula still helps prevent a low-clearance employee from reading higher-classified records through the authorised system. However, it is irrelevant once an insider legitimately obtains ciphertext and also possesses the key, or copies plaintext after authorised decryption. Access control cannot make a compromised key secret again.

C. Adaptation or Redesign (2 Marks)

Decision: MODIFY the educational design rather than retain the same network-wide key. Give each clinic a separate key, use a fresh session key for each exchange, rotate keys on a schedule and immediately after suspected compromise, and remove predictable fixed headers from the encrypted body. Add random padding before encryption to make message lengths and common formats less obvious. If a report must be hidden in an image, encrypt the report first and then embed the ciphertext in a separate non-diagnostic image copy.

For a Vigenère key of length m, the nominal key space is 26^m. Using a different key per clinic and per session increases the number of independently managed keys, but it does not repair the structural weakness of a repeated-key classical cipher. Random padding reduces format clues but does not create modern cryptographic security. More keys also increase distribution, backup, revocation and staff-support costs.

D. Evidence-Based Defence (2 Marks)

The evidence shows why modification is necessary: Caesar and monoalphabetic substitution preserve letter frequencies; transposition preserves the frequency counts; repeated-key Vigenère exposes periodic patterns; and Hill is linear and vulnerable to known-plaintext recovery. The mathematical size of a key space alone cannot protect the data if the cipher leaks structure or the key is copied.

The modified plan limits the damage of one clinic's compromise because other clinics use different keys. Session keys and rotation reduce the amount of ciphertext exposed under one key. Random padding and removal of fixed headers reduce known-format clues. The trade-off is greater key-management effort, so the central hospital must provide simple automated key distribution and clear procedures for rural staff.

Final judgement: classical ciphers cannot guarantee modern confidentiality, integrity, authentication, non-repudiation, forward secrecy or safe key management for patient data. They are useful for demonstrating cryptographic principles, but a real healthcare network must use modern authenticated encryption and a managed identity/key infrastructure. The medical image must remain diagnostically unchanged; any hidden report belongs in a separate copy with validation and a recoverable original.

Overall Conclusion

A secure healthcare exchange system must protect both patient privacy and patient safety. Risk analysis identifies insiders, prescription modification and impersonation as serious threats. Bell-LaPadula provides a useful confidentiality policy, but it is too rigid for emergencies and does not guarantee integrity. Classical ciphers demonstrate substitution, transposition, key spaces and cryptanalysis, but they are vulnerable to statistical and known-plaintext attacks. The final design therefore uses per-clinic and per-session key separation as an educational improvement, strict access control, auditing and controlled emergency access, while clearly recognising that real patient records require modern authenticated encryption and secure key management.

MARK DISTRIBUTION

Task	**Marks**
Task 1 – Threats, Attacks and Risk Evaluation	4
Task 2 – Security Goals Mapping	3
Task 3 – Bell-LaPadula Application and Critique	5
Task 4 – Cipher Design	4
Task 5 – Mathematical Analysis	5
Task 6 – Cryptanalytic Resistance	4
Task 7 – Transposition / Product Ciphers	3
Task 8 – Steganography Evaluation	3
Task 9 – Implementation & Testing	4
Task 10 – Experimental Evidence	3
Task 11 – Final Design Recommendation	4
Subtotal	**42**

Task 12 – Day-4 Design Challenge	**8**
TOTAL	**50**